



特徵選擇與維度縮減 Feature Selection & Dimension Reduction

劉益宏

國立陽明交通大學電機系、電控所 教授

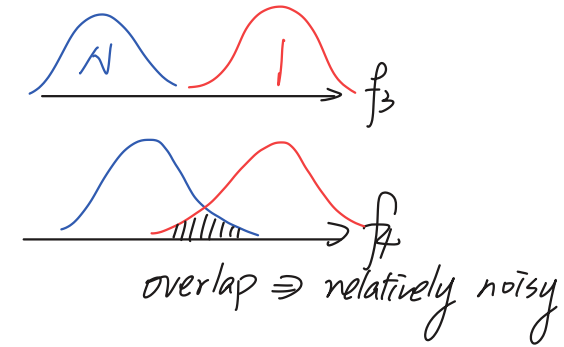
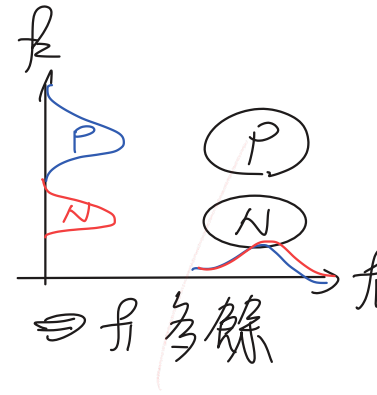
yhliu182@nycu.edu.tw



Feature selection

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- **Goal** : given a set of features, $f_1, f_2, \dots, f_n$, to determine the best or optimal features among the set.
- **Reason** : Some features may be **noisy** and/or **redundant**. These features might drop the classification or prediction accuracy. **The best way is to remove them from the given feature set**, and preserve the ones that can give better classification or prediction accuracy.
- **Why feature selection algorithms?** More efficient and reliable.



→ to evaluate the feature "systematically"



What are good features?

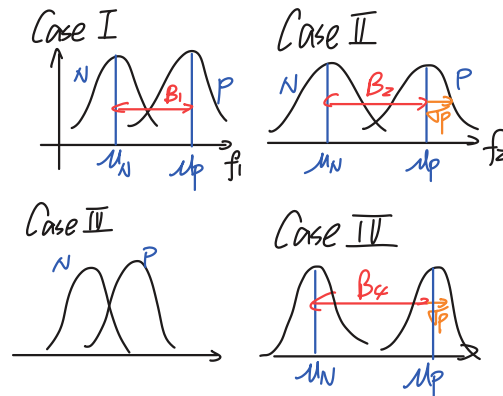
- Two criteria to evaluate a feature :

- Between-class separability 類別之間的分離度
- Within-class variation $\max |u_P - u_N|$
 $\min \sigma_P + \sigma_N$

- A good feature should be capable of providing a **large between-class separability** (= between-class scatter) and a **small within-class variation** (= within-class scatter or **intra-class inertia**). 慣性

內
inter 之間

$\frac{1}{\sigma}$ $\uparrow$ 即 $\sigma \uparrow \Rightarrow \downarrow$



$$B_2 = B_4 > B_1$$

• f_2, f_4 有 largest between classification distance

• f_4 的 $\sigma(50) < f_2$ 的 σ

↳ within class variation

⇒ 特徵的不變形

↓ max between class separability
min within class variation



Feature selection (FS)

1. Filter-based feature selection

- Filter methods use pre-defined measures to rank the features without using any classifiers to calculate the classification accuracy.
- e.g., Fisher's method.

计算 F score 不需 classifier

2. Wrapper-based feature selection

- Wrapper methods determine the optimal feature combination **directly** by using the **generalization performance**. ✓
- e.g., Sequential forward selection (SFS), sequential backward selection (SBS). → both are heuristic search method.



Fisher's criterion (費雪法) for feature selection (1/3)

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- Given the following: $\mathbf{x}_{ij} \in R^d$ is the j th training data of the i th class, $i = 1, \dots, c$, $j = 1, \dots, n_i$, where n_i is the number of the training data of class i , $\mathbf{m}_i$ is the mean of the training data of the i th class, and $\mathbf{m}$ is the mean of the training data from all classes (overall mean). Let $P_i = n_i / \sum_{j=1}^c n_j$ be the prior probability of the i th class.

- Within-class scatter matrix $\mathbf{S}_w$ is

$$\mathbf{S}_w = \sum_{i=1}^c P_i \left(\frac{1}{n_i} \sum_{j=1}^{n_i} (\mathbf{x}_{ij} - \mathbf{m}_i) (\mathbf{x}_{ij} - \mathbf{m}_i)^t \right)$$

Covariance
去中心化
 $d \times d$

- Between-class scatter matrix $\mathbf{S}_b$ is

$$\mathbf{S}_b = \sum_{i=1}^c n_i (\mathbf{m}_i - \mathbf{m})(\mathbf{m}_i - \mathbf{m})^t$$

$d \times d$

$$\mathbf{S}_D = \sum_{i=1}^c P_i \left(\frac{1}{n_i} \sum_{j=1}^{n_i} (\mathbf{x}_{ij} - \mathbf{m}_i)(\mathbf{x}_{ij} - \mathbf{m}_i)^t \right)$$

$$\left. \begin{array}{l} \mathbf{S}_w = d \times d \\ \mathbf{S}_b = d \times d \end{array} \right\}$$



Fisher's criterion (費雪法) for feature selection (2/3)

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- The Fisher's score (F-score) of the k th feature is given by

$$F(k) = \frac{S_b(k)}{S_w(k)}, \quad k = 1, 2, \dots, d$$

the larger, the better

↖ kth 對角值 ↘ kth 對角值

where $S_b(k)$ and $S_w(k)$ are the k th diagonal elements of the matrices S_b and S_w , respectively.

- Fisher's class separability criterion.** the Fisher's method calculates a Fisher score (F-score) for each feature. A higher F-score corresponds to a higher ratio of the between-class separability to the within-class variation. Thus, a feature with a higher Fisher's score corresponds to a more distinguishing feature in classification.
- The higher the Fisher's score is, the better the feature is.

Q: Assume f_1, f_2, f_3, f_4

and F-score(f_1) = 0.8

F-score(f_2) = >

F-score(f_3) = 1.2

F-score(f_4) = 0.3

⇒ Optimal features



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Fisher's criterion (費雪法) for feature selection (3/3)

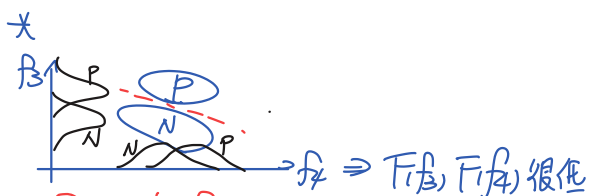
Feature selection based on Fisher's criterion and top- n -F-score-ranked feature selection strategy.

Assume $M_f = 10^4$

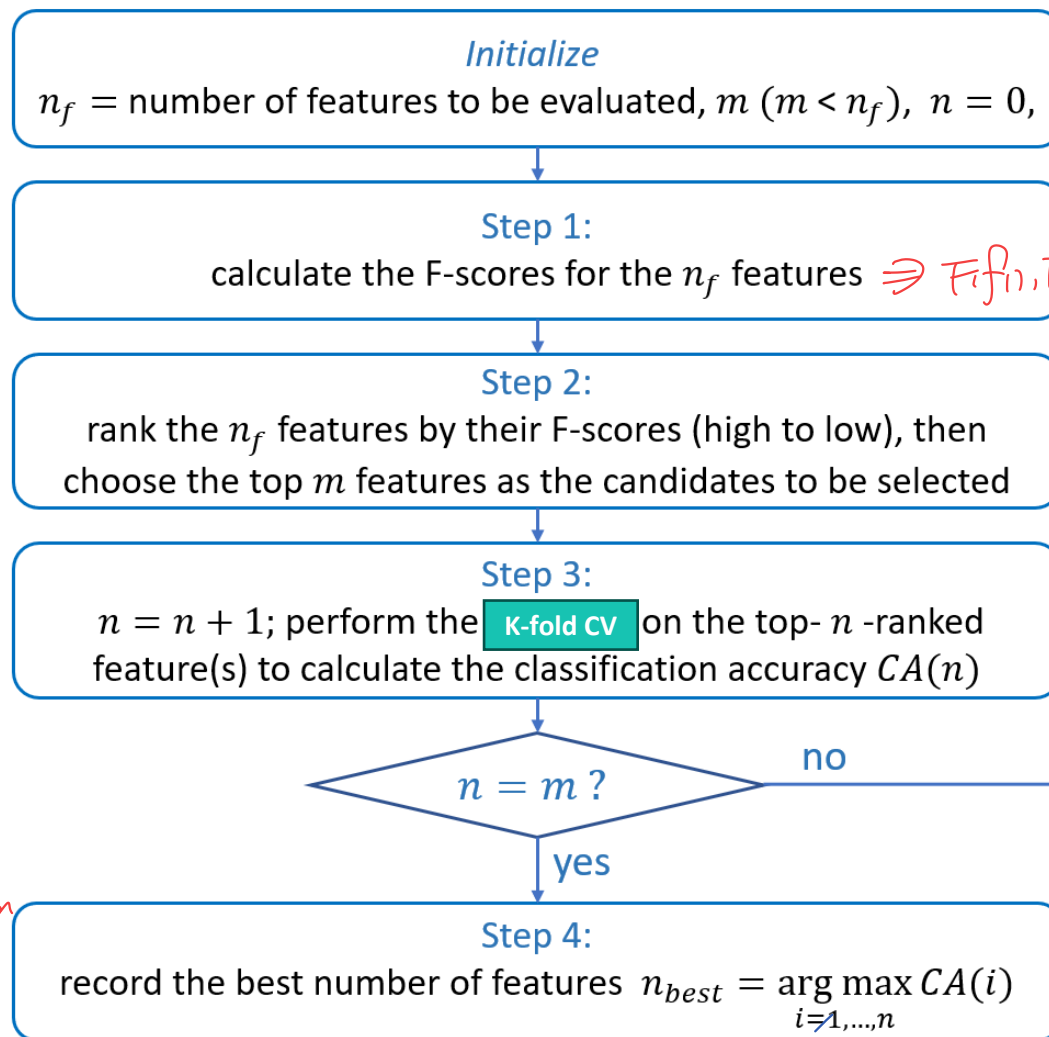
train data = 10^3

⇒ 合理的最佳特徵數量應要小於 10^3

⇒ 故 search range 應定在 $m=600$



⇒ 僅兩 weak feature combination result in a good classification



* 特徵數目最好
不要大於訓練數目

$f_{(1)}, f_{(2)}, \dots, f_{(m)}$
top feature



$f_{(1)}, f_{(2)}, \dots, f_{(m)}$

① only $f_{(1)} \Rightarrow$ k-fold CV $\Rightarrow$ Classification Accuracy $CA(1)$

② $f_{(1)}, f_{(2)} \Rightarrow$ k-fold CV $\Rightarrow CA(2)$
top 2 feature

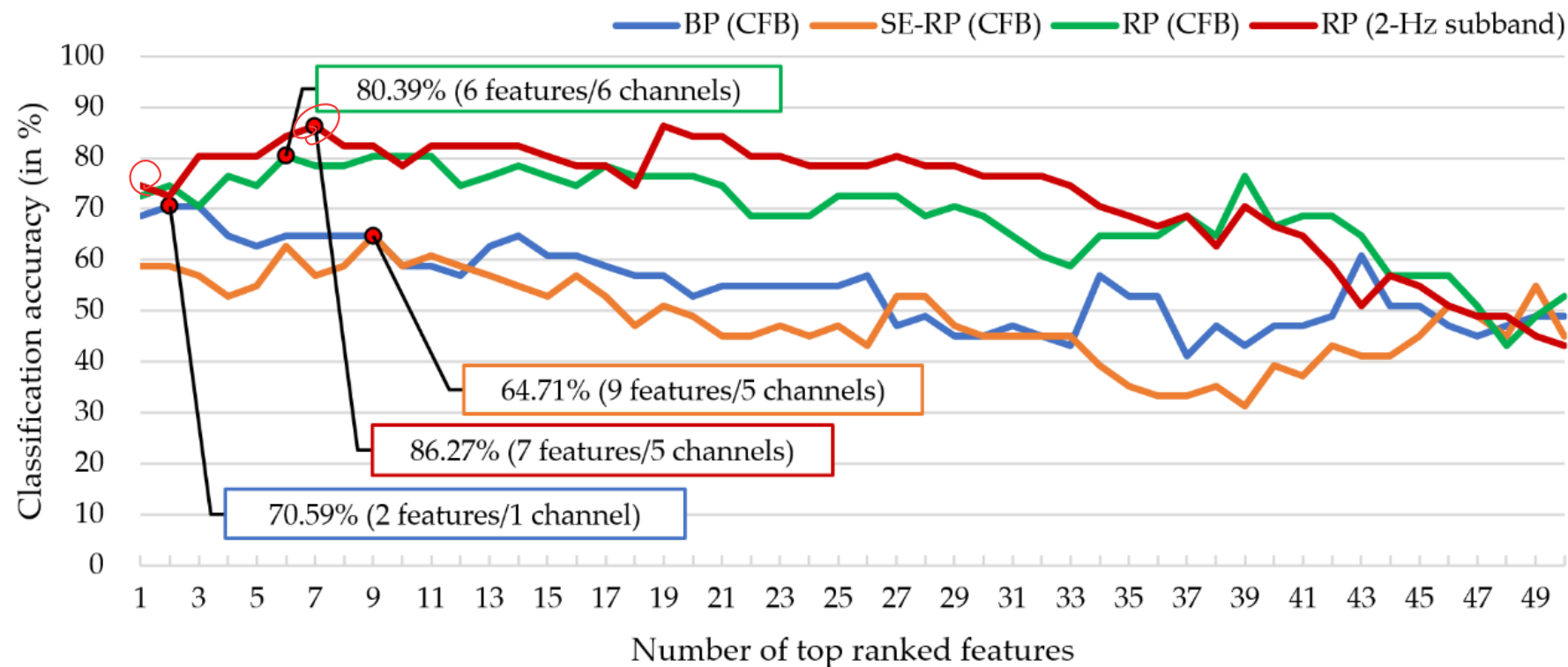
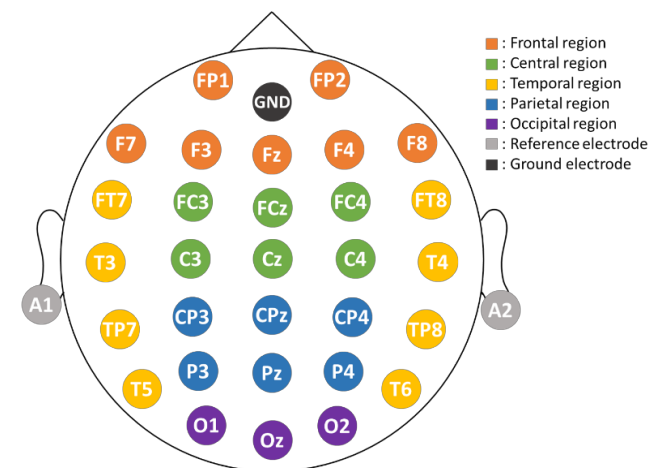
$f_{(1)}, f_{(2)}, \dots, f_{(m)} \Rightarrow$ k-fold CV $\Rightarrow CA(m)$

the best num of optional feature





Example. EEG feature selection

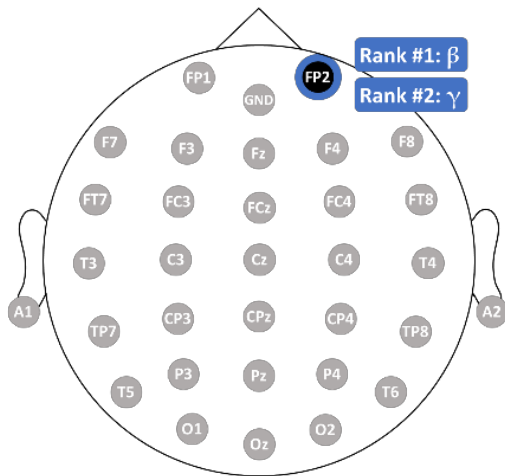


Comparison of MCI-HC classification accuracy among conventional frequency band (CFB)-based BP, SE-RP (CFB), RP (CFB), and 2-Hz-width subband-based RP based on the LDA classifier. The total numbers of feature candidates for the four features are 150, 150, 2175, and 9135, respectively. The best classification accuracies for the four features are 70.59%, 64.71%, 80.39%, and 86.27%, respectively. In terms of RP (2-Hz subband), its accuracy curve reached the maximum 86.27% when the number of top-F-score-ranked-feature increased to 7, which occupy only a very small portion of the whole RP feature candidates ($7/9135 = 0.07\%$).

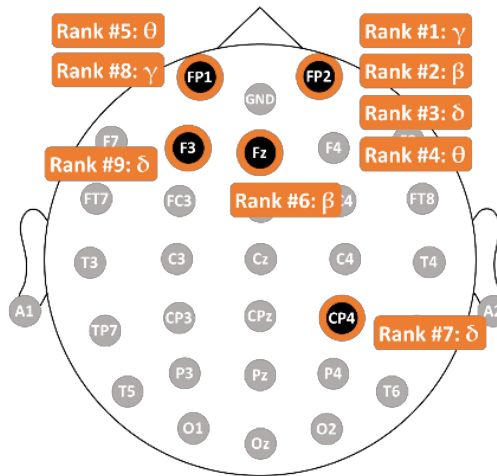


Example.

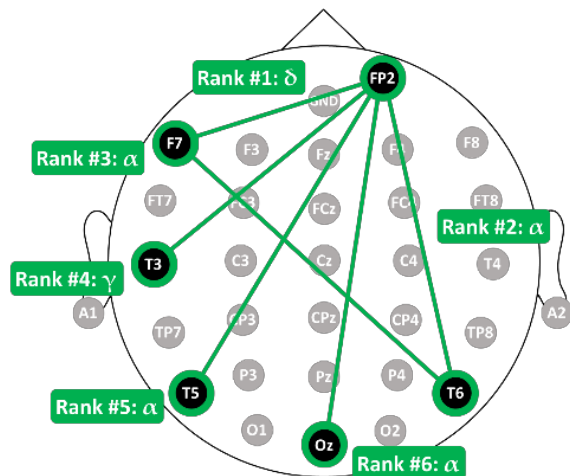
Best BP (CFB) channels



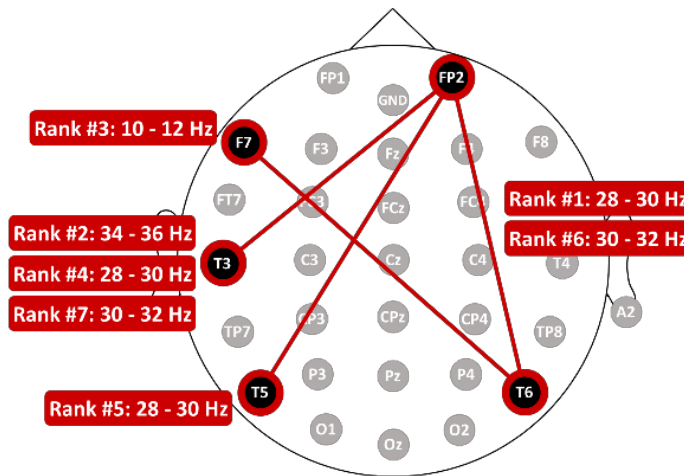
Best SE-RP (CFB) channels



Best RP (CFB) channel pairs



Best RP (2-Hz subband) channel pairs



The best feature subsets of the BP (CFB), SE-RP (CFB), RP (CFB), and RP (2-Hz subband) features. **Upper-left:** the best set for BP includes beta (top-1-ranked) and gamma (top-2-ranked) BPs, and their electrode locations are at FP2. **Upper-right:** the best set of SE-RP includes 9 features at 5 different electrode channels. **Bottom-left:** the best set of RP (CFB) includes 6 features from 6 different electrodes. **Bottom-right:** the best set of RP includes 7 features from 5 different electrodes.



Remarks on Fisher's method for FS

- Advantages :
 1. Easy to implement
 2. Fast – rank the features directly
 3. Suitable for choosing a set of "feature candidates" among a large set of features → filtering
- Disadvantages
 1. Does not consider the interaction between features (the joint probability)
 2. Easy to ignore the weak features which may contribute to high classification accuracy.



Sequential Forward Selection (SFS): 序列前向選擇

Algorithm SFS

- Initialization : N_s = number of feature candidates
- Step 1.** finds the **best single feature** which gives the highest k -fold cross validation (k -fold CV) classification accuracy.
- Step 2.** Subsequently, $N_s - 1$ **pairs** of features are formed by combining each of the remaining features with the best single feature, and **the best pair** (i.e., the pair which gives the highest k -fold CV classification accuracy) is selected.
- Step 3.** Following the same logic, $N_s - 2$ **triples** of features are formed using every of the remaining features and the best feature pair, and the best triple is selected.
- Step 4.** This procedure is repeated until all the N_s features are tested. Finally, the best feature set is the one resulting in the highest k -fold CV classification accuracy. The top- n -ranked features giving the highest k -fold CV classification accuracy form the **optimal feature subset**, where $1 \leq n \leq N_s$.

Given f_1, f_2, f_3, f_4

$k=4$

Step 1 $CA(f_1)$, $CA(f_2)$, $CA(f_3)$, $CA(f_4)$
 80% 85% 75% 77%
 best

Step 2 其餘跟 f_{best} 搭配找最好
 $CA(f_1, f_2)$, $CA(f_3, f_2)$, $CA(f_4, f_2)$
 87% 89% 85%
 best pair

Step 3 $CA(f_2, f_3, f_1)$ $CA(f_2, f_3, f_4)$
 91% 95%
 best triple

Step 4 $CA(f_1, f_2, f_3, f_4)$
96%

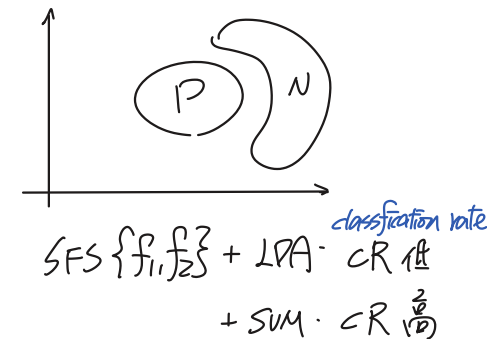


SFS remarks

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- SFS: a wrapper-based **greedy** approach for feature selection, which has the advantages of achieving better accuracy than filter-based feature selection approach (Guyon, 2003).
- **Wrapper approaches include the interaction between feature subset and classification model** (Saeys et al., 2007). In other words, wrapper-based methods are classifier-specific in which the methods search for the best subset of features that optimizes the generalization classification accuracy of a chosen classifier (Kudo et al., 2000)
- e.g., for the classifiers LDA and SVM, the optimal feature subsets selected by the SFS method could be different.

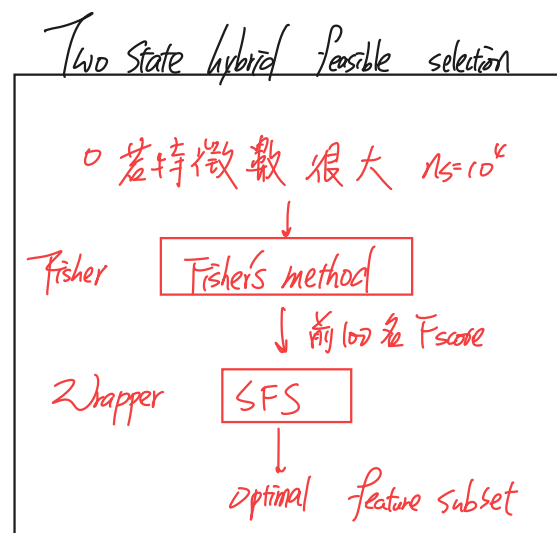
⇒ time-consuming if N is large ' 耗耗時 '



⇒ 不同 classification 会造成不同 CR
⇒ 即便 result 与特定分类器相关, 不通用

joint

SFS 考虑不同 feature 之间的可能性
考虑特征跟分类器的互动





SFS-based feature selection and the grid-search-based SVM parameter optimization procedure

Nes

Initialize: $N_g = 961$ (number of SVM parameter grids(C, γ))
 N_s = total number of the features to be evaluated

Step 1

Perform the **grid search**: choose an SVM parameter grid (C, γ)

Step 2

Perform feature selection based on **SFS** and k -fold cross validation (CV)

no

All the N_g parameter grids
are tested ?

yes

Step 3

Record the optimal feature subset, optimal parameters, and the corresponding (the highest) k -fold CV classification accuracy

- 1) Record the highest k -fold CV classification accuracy, i.e., $\max_{(C, \gamma)} CA(n^*)$.
- 2) Optimal SVM parameters are $(C^*, \gamma^*) = \arg \max_{(C, \gamma)} CA(n^*)$.
- 3) Optimal features are the top- n^* features determined by the SVM classifier with the optimal hyperparameter (C^*, γ^*) .

Feature Selection based on SFS and k -fold CV

Given the chosen parameter grid (C, γ):

- 1) **Step 1: Find the best single feature among the N_s feature candidates.** Find the best single BRS feature which gives the highest k -fold CV classification accuracy, denoted as $CA(1)$.
- 2) **Step 2: Determine the best pair of the features.** $N_s - 1$ pairs of the features are formed by combining each of the remaining features with the best single feature found in Step 1, and the best pair is the one (top-2 features) that gives the highest k -fold CV classification accuracy, denoted as $CA(2)$.
- 3) **Step 3: Determine the best triple of the features.** $N_s - 2$ triples of features are formed using every of the remaining features and the best pair found in Step 2, and the best triple that gives the highest classification k -fold CV accuracy (denoted as $CA(3)$) is selected.
- 4) **Step 4. This procedure is repeated until all the N_s features are tested.** Then, record $CA(n)$, $n = 1, \dots, N_s$
- 5) **Step 5. Determine the optimal feature subset.** Top- n^* -ranked features giving the highest k -fold CV accuracy form the optimal feature subset, where $n^* = \arg \max_n CA(n)$.



Sequential backward selection (SBS): 序列倒向選擇

Algorithm SBS

1. Let N represent the number of feature candidates.
2. **Step 1.** Initially, a k -fold CV is performed on the dataset with all the N features.
3. **Step 2.** Following that, each feature is removed, one at a time. The k -fold CV is applied to all $N-1$ features subsets. The worst feature is discarded. The worst feature means that without it, the subset with the remaining $N-1$ features gives the highest cross-validation accuracy among all the $N-1$ subsets.
4. **Step 3.** In the next round, each of the $N-1$ features is deleted one at a time, and the worst feature is discarded, using the same logic. After that, a subset with $N-2$ features is formed.
5. **Step 4.** This procedure is repeated until none of the N features is left. Finally, the optimal feature subset is the one that produces the highest 5-fold CV accuracy.

k -fold

Given f_1, f_2, f_3, f_4 . $N=4$

Step 1 $CA(f_1, f_2, f_3, f_4) = 85\%$ as baseline

Step 2 $CA(f_1, f_2, f_3)$, $CA(f_1, f_2, f_4)$, $CA(f_1, f_3, f_4)$, $CA(f_2, f_3, f_4)$

87%

$\Rightarrow f_4$ 不好

$\Rightarrow$ 拿掉 f_4

79%

82%

80%

$\Rightarrow f_1$ 很重要



讓每一個 feature 都有機會參與

$\Rightarrow$ 通常比 SFS 好

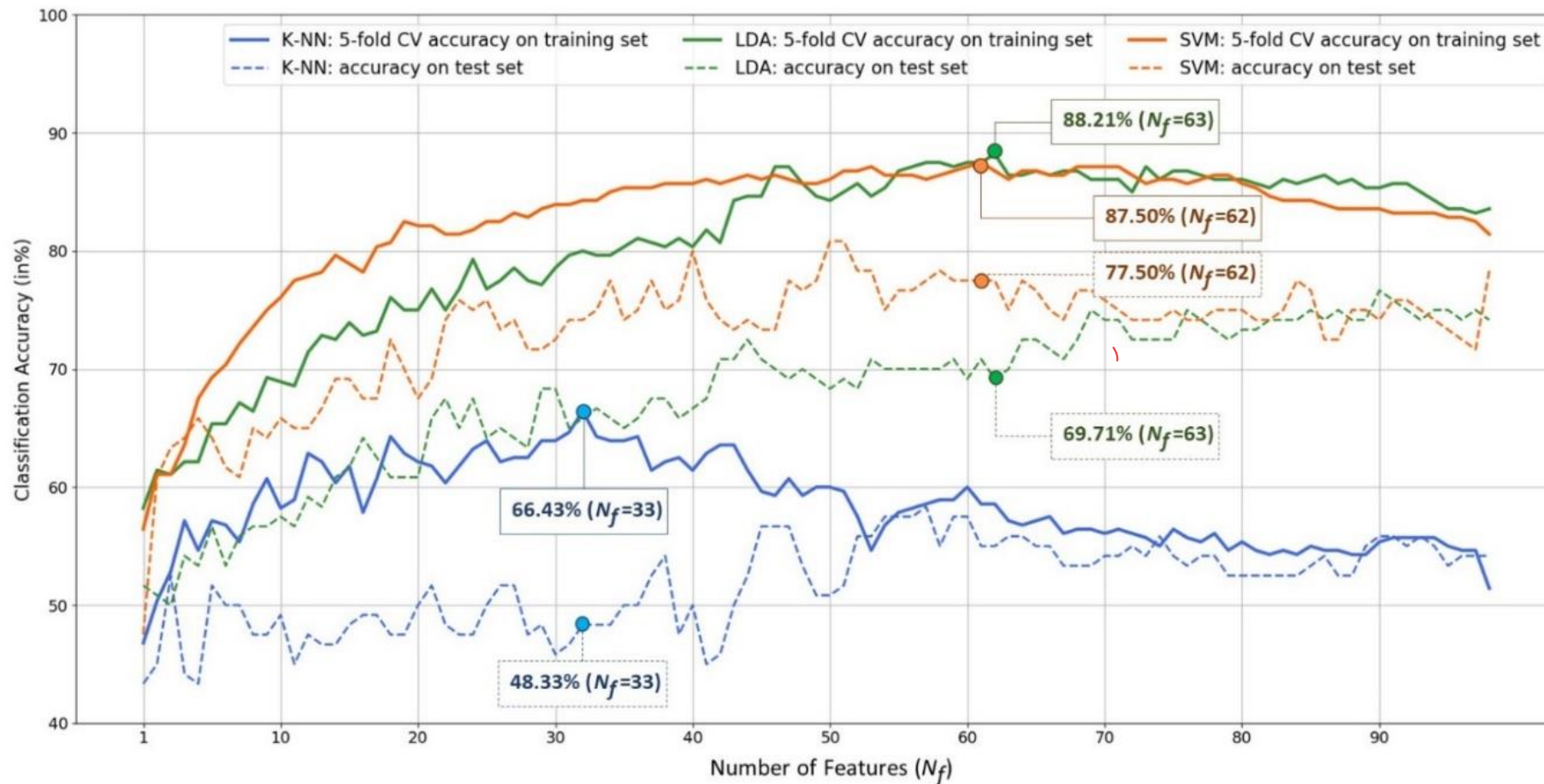
$\Rightarrow$ 因為讓弱的也會參與到



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An example of using the SBS method to determine the optimal feature subsets for different classifiers

決定分類器



交叉驗證的出來解(最高點),不一定是test的最高點



Remarks on SBS

- **Limitation.** It is unable to re-evaluate the utility of the features that have been discarded. In other words, the optimal feature subset selected by the SBS algorithm may not really be the optimal.
- **Solution.** Other greedy and randomized feature selection methods, such as the genetic algorithm (GA), can be used to overcome this limitation.
- **Notice.** The optimal feature subset determined by SBS may differ from those identified by other methods like GA or SFS.



維度縮減 (Dimension Reduction)

1. 線性子空間法 Linear subspace method.

- 主成分分析 Principal component analysis (PCA)
- 費雪線性鑑別分析 Fisher's linear discriminant analysis (FLDA)

2. 非線性子空間法 Nonlinear subspace method.

- 核主成分分析 **Kernel** PCA (1998)
- 廣義鑑別分析 Generalized discriminant analysis (**GDA**) (2000) → also known as *kernel LDA*

3. 流形學習法 **Manifold learning methods**

基底數目 $\propto$ 維度數量

- 局部線性嵌入 Locally linear embedding (LLE)
- t -分布隨機鄰近嵌入法 t -distributed stochastic neighbor embedding (t -SNE)

4. 神經網路 Neural networks

- 自組織映射 (self-organizing map, SOM)

$$Def = \mathbb{R}^d$$

將高維向量映射至低維向量

$$f = \text{function} \cdot \vec{x} \in \mathbb{R}^d \rightarrow \vec{y} \in \mathbb{R}^n$$

$$\text{且 } n \ll d$$

$\Rightarrow f$ perform dimension reduction task on x

$$\Rightarrow \vec{y} = W \vec{x} \Rightarrow \mathbb{R}^n \text{ 乘 } W$$

$$n \times n \quad n \times d \quad d \times n$$

* 若在二維中存在多點於同一平面

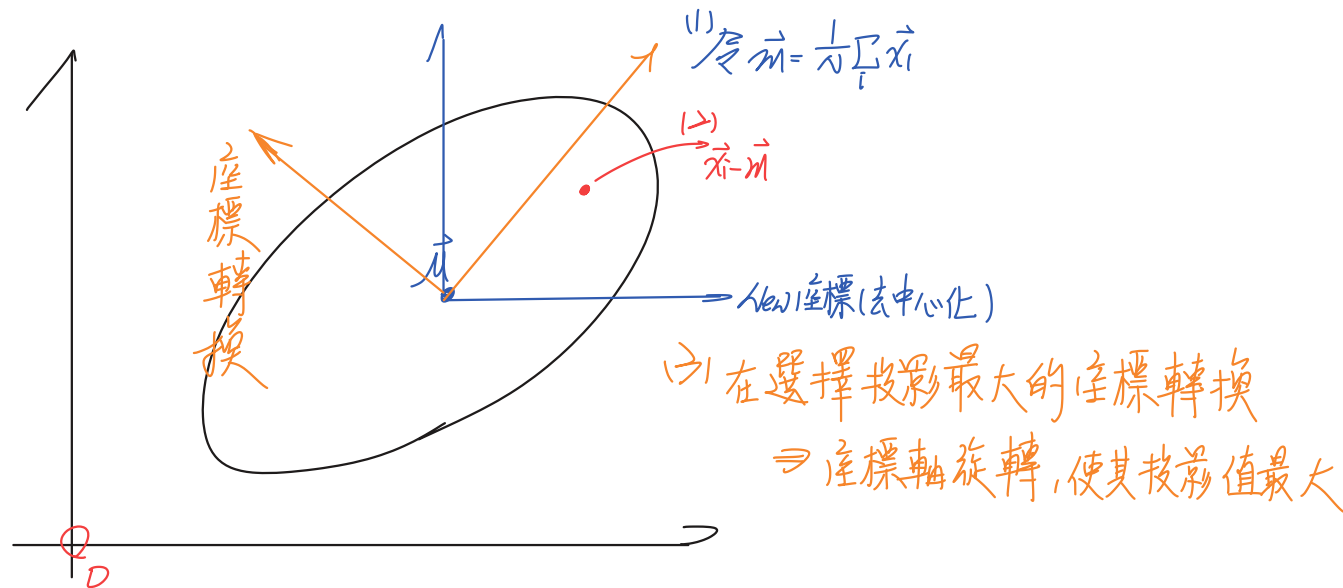
則此平面為 manifold, 為二維的子空間



主成分分析 (principal component analysis, PCA) 演算法

1. PCA利用(1) 中心化、(2) 找新座標、(3) 座標轉換來進行維度縮減
2. PCA是一種在原空間內進行維度縮減的線性轉換法

概念介紹：



$$S = \left\{ \vec{x}_i \in \mathbb{R}^d \right\}_{i=1}^N \text{ distribution}$$



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主成分分析 (principal component analysis, PCA) 演算法

去中心化後投影

principal component analysis

- Given a set of **training data** $\mathbf{x}_i \in R^d, i=1, \dots, m$, PCA solves the eigenvalue problem as

$$\mathbf{C}\mathbf{v} = \lambda\mathbf{v} \Rightarrow \text{得到 } \lambda \text{ 的 eigenvalue 對應到 } \vec{v} \text{ 的 eigenvectors}$$

where **C** is the $d \times d$ **covariance matrix** of the **training data**, and **v** is the eigenvector of **C**, and λ is the eigenvalue of **v**.

- For a **test data** (a vector) $\mathbf{x} \in R^d$, its projections onto the orthonormal eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ associated with the n largest nonzero eigenvalues form a new vector $\mathbf{y} \in R^n$

$$\mathbf{y} = [\mathbf{v}_1 \ \mathbf{v}_2 \ \dots \ \mathbf{v}_n]^T (\mathbf{x} - \mathbf{m}) = \begin{bmatrix} \vec{v}_1^T (\vec{x} - \vec{m}) \\ \vec{v}_2^T (\vec{x} - \vec{m}) \\ \vdots \\ \vec{v}_n^T (\vec{x} - \vec{m}) \end{bmatrix}$$

平移至新座標軸

where **m** is the mean vector of the training data.

$\Rightarrow$ 不須要 label $\Rightarrow$ unsupervised learning

$$C = \frac{1}{m-1} \sum_{i=1}^m \frac{(\vec{x}_i - \vec{m})}{d \times 1} \frac{(\vec{x}_i - \vec{m})^T}{1 \times d}$$

若樣本數夠大 $\frac{1}{m-1} \rightarrow \frac{1}{m}$
(因為我們是用抽樣)

1) 從 d 個 $\vec{v}_i$ 中選 n 個
choose the n eigenvalue associated with the largest $\lambda_i \Rightarrow 1 \ll d \ll R$
 $\Rightarrow$ 選出 $\vec{v}_i$ 最大的 n 個

2) 正規化

$$\vec{v}_i = \frac{\vec{v}_i}{\|\vec{v}_i\|}, i=1, \dots, n$$

3) 排列

finally, the transformation matrix $W = [\vec{v}_1 \ \vec{v}_2 \ \dots \ \vec{v}_n]$ training end



PCA手算例 (1/5)

助教 官韋宏製作

Step 1

S : training set, $S = \{(\mathbf{x}_i, y_i)\}_{i=1, \dots, 4}$ ^{4筆 training data}

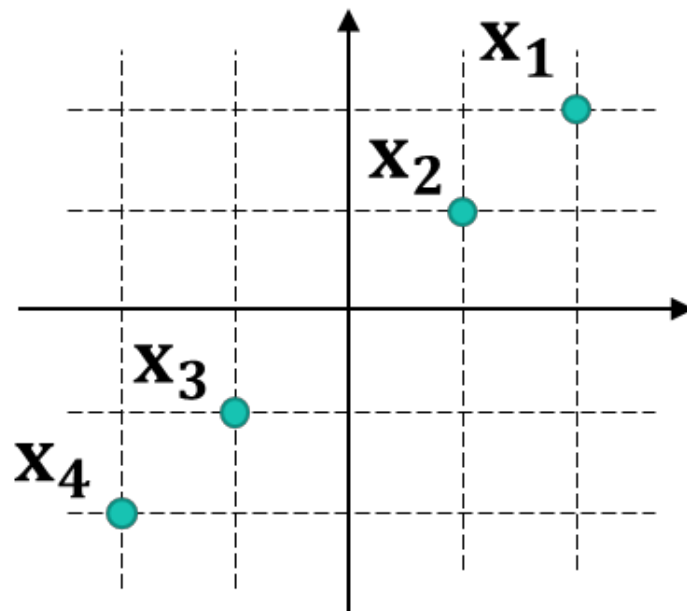
$$\mathbf{x}_1 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}; \mathbf{x}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}; \mathbf{x}_3 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}; \mathbf{x}_4 = \begin{bmatrix} -2 \\ -2 \end{bmatrix} \quad \text{原空間 dim}=2$$

Step 2

Calculate the covariance matrix of $\mathbf{S}$:

$$\begin{aligned} \mathbf{C} &= \frac{1}{4-1} \sum_{i=1}^4 (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T \\ &= \frac{1}{3} \left(\begin{bmatrix} 2 \\ 2 \end{bmatrix} \begin{bmatrix} 2 & 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} + \begin{bmatrix} -1 \\ -1 \end{bmatrix} \begin{bmatrix} -1 & -1 \end{bmatrix} + \begin{bmatrix} -2 \\ -2 \end{bmatrix} \begin{bmatrix} -2 & -2 \end{bmatrix} \right) \\ &= \frac{1}{3} \begin{bmatrix} 10 & 10 \\ 10 & 10 \end{bmatrix} \end{aligned}$$

$C=2 \times 2$ 是 training data 的維度相乘
跟 data 數無關





PCA手算例 (2/5)

The eigenvalue problem can be written as :

$$\mathbf{C}\mathbf{v} = \lambda\mathbf{v}$$

$$\Rightarrow (\lambda\mathbf{I}_{2 \times 2} - \mathbf{C})\mathbf{v} = \mathbf{0}$$

$$\Leftrightarrow (\lambda\mathbf{I} - \mathbf{C}) : \text{singular}$$

$$\Leftrightarrow |\lambda\mathbf{I} - \mathbf{C}| = 0 \dots\dots \text{特徵方程式}$$

$\mathbf{I}_{2 \times 2}$: 2×2 之單位矩陣

$\mathbf{0}$: 零向量

$| \quad |$: 行列式 (determinant)

$\mathbf{v} \in \mathbb{R}^{2 \times 1}$

$$\lambda\mathbf{I} - \mathbf{C} = \begin{bmatrix} \lambda - \frac{10}{3} & -\frac{10}{3} \\ -\frac{10}{3} & \lambda - \frac{10}{3} \end{bmatrix}$$

$$\det(\lambda\mathbf{I} - \mathbf{C}) = \left(\lambda - \frac{10}{3}\right)^2 - \frac{100}{9}$$

$$\Rightarrow \lambda^2 - \frac{10}{3}\lambda + \frac{100}{9} - \frac{100}{9} = 0$$

$$\Rightarrow \lambda \left(\lambda - \frac{20}{3}\right) = 0$$

$$\Rightarrow \lambda = \frac{20}{3}, 0 \rightarrow \text{相異實根}$$



PCA手算例 (3/5)

當 $\lambda = 20/3$

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$\left(\frac{20}{3}\mathbf{I} - \mathbf{C}\right)\mathbf{v} = \mathbf{0}$$

$$\Rightarrow \begin{bmatrix} 10/3 & -10/3 \\ -10/3 & 10/3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\Rightarrow v_1 = 1, v_2 = 1$ 是一組解

$\therefore \mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ 是相對於 $\lambda = \frac{20}{3}$ 的一個特徵向量

當 $\lambda = 0$

$$(\mathbf{0I} - \mathbf{C})\mathbf{v} = \mathbf{0}$$

$$\Rightarrow \begin{bmatrix} -10/3 & -10/3 \\ -10/3 & -10/3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\Rightarrow v_1 = -1, v_2 = 1$ 是一組解

$\therefore \mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ 是相對於 $\lambda = 0$ 的一個特徵向量

Note

$$\langle \mathbf{v}_1, \mathbf{v}_2 \rangle = \mathbf{v}_1 \cdot \mathbf{v}_2 = \mathbf{v}_1^T \mathbf{v}_2 = 0$$

$\therefore \mathbf{v}_1, \mathbf{v}_2$: Orthogonal 正交



PCA手算例 (4/5)

* Cov : 對稱矩陣 保證正交

Normalization :

$$\mathbf{v}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \quad \mathbf{v}_2 = \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|} = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

$\therefore \mathbf{v}_1, \mathbf{v}_2$: Orthonormal 單範正交

Next, project the 4 data onto the two eigenvectors, respectively

① 將 $\mathbf{x}_1$ 投影在 $\mathbf{v}_1$ 上 :

$$\begin{aligned} & \mathbf{v}_1 \cdot (\mathbf{x}_1 - \bar{\mathbf{x}}) \\ &= \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 2 \end{bmatrix} \\ &= \frac{4}{\sqrt{2}} = 2\sqrt{2} \end{aligned}$$

將 $\mathbf{x}_1$ 投影在 $\mathbf{v}_2$ 上 :

$$\begin{aligned} & \mathbf{v}_2 \cdot (\mathbf{x}_1 - \bar{\mathbf{x}}) \\ &= \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 2 \end{bmatrix} \\ &= 0 \end{aligned}$$



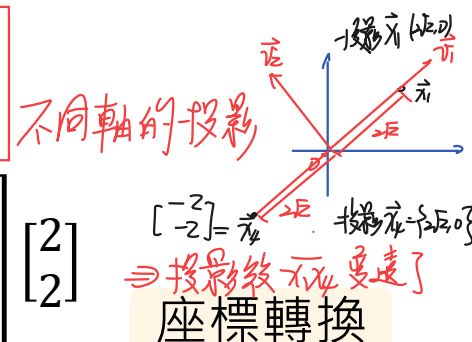
若令 $\mathbf{W} = [\mathbf{v}_1 \quad \mathbf{v}_2]_{2 \times 2}$ 為 transformation matrix

$$\text{則 } \mathbf{W}^T (\mathbf{x}_1 - \bar{\mathbf{x}}) = \begin{bmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \end{bmatrix} (\mathbf{x}_1 - \bar{\mathbf{x}})$$

mean

$$= \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 2\sqrt{2} \\ 0 \end{bmatrix} \triangleq \mathbf{z}_1$$



$\mathbf{z}_1$ 是 $\mathbf{x}_1$ 在以 $\mathbf{v}_1, \mathbf{v}_2$ 為座標軸下的新座標向量



PCA手算例 (5/5)

② 將 $\mathbf{x}_2$ 投影在 $\mathbf{v}_1$ 和 $\mathbf{v}_2$ 上：

$$\mathbf{W}^T(\mathbf{x}_2 - \bar{\mathbf{x}}) = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} \sqrt{2} \\ 0 \end{bmatrix} \triangleq \mathbf{z}_2$$

$$\begin{aligned} \vec{z}_1 &= \mathbf{W}^T(\vec{x}_1 - \vec{m}) \\ \vec{z}_2 &= \mathbf{W}^T(\vec{x}_2 - \vec{m}) \\ \vec{z}_3 &= \mathbf{W}^T(\vec{x}_3 - \vec{m}) \\ \vec{z}_4 &= \mathbf{W}^T(\vec{x}_4 - \vec{m}) \end{aligned} \left. \begin{array}{l} \\ \\ \\ \end{array} \right\} \begin{array}{l} \text{可以看出第3項} \\ \text{皆為0} \end{array}$$

② 將 $\mathbf{x}_3$ 投影在 $\mathbf{v}_1$ 和 $\mathbf{v}_2$ 上：

$$\mathbf{W}^T(\mathbf{x}_3 - \bar{\mathbf{x}}) = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} -1 \\ -1 \end{bmatrix} = \begin{bmatrix} -\sqrt{2} \\ 0 \end{bmatrix} \triangleq \mathbf{z}_3$$

② 將 $\mathbf{x}_4$ 投影在 $\mathbf{v}_1$ 和 $\mathbf{v}_2$ 上：

$$\mathbf{W}^T(\mathbf{x}_4 - \bar{\mathbf{x}}) = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} -2 \\ -2 \end{bmatrix} = \begin{bmatrix} -2\sqrt{2} \\ 0 \end{bmatrix} \triangleq \mathbf{z}_4$$



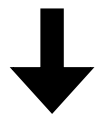
PCA手算例 (結論)

data 投影在 Covariance eigenvector 上的值 $\triangleq$ principal component (PC)

Summary

$$\begin{array}{c} \mathbf{x}_1 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}; \mathbf{x}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}; \mathbf{x}_3 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}; \mathbf{x}_4 = \begin{bmatrix} -2 \\ -2 \end{bmatrix} \\ \text{PCA} \downarrow \\ \mathbf{z}_1 = \begin{bmatrix} 2\sqrt{2} \\ 0 \end{bmatrix}; \mathbf{z}_2 = \begin{bmatrix} \sqrt{2} \\ 0 \end{bmatrix}; \mathbf{z}_3 = \begin{bmatrix} -\sqrt{2} \\ 0 \end{bmatrix}; \mathbf{z}_4 = \begin{bmatrix} -2\sqrt{2} \\ 0 \end{bmatrix} \end{array}$$

- 若取第一個主成份 $\Leftrightarrow$ 取在 $\mathbf{v}_1$ 之投影 $\Rightarrow$ 取 λ 值比較大的 eigenvector 上的投影
- $\because$ 在 $\mathbf{v}_2$ 上投影皆為 0 $\Rightarrow$ 此特徵值對分類沒幫助



$$z_1 = 2\sqrt{2}; z_2 = \sqrt{2}; z_3 = -\sqrt{2}; z_4 = -2\sqrt{2}$$

取 $\vec{v}_1$ 當作投影則 $w = \vec{v}_1$

$$\therefore z = \vec{w}^T (\vec{x} - \vec{m})$$

$$\Rightarrow z = \vec{v}_1^T (\vec{x} - \vec{m}), z = \text{scalar}$$

$$\mathbf{x}_1 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}; \mathbf{x}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}; \mathbf{x}_3 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}; \mathbf{x}_4 = \begin{bmatrix} -2 \\ -2 \end{bmatrix} \quad \text{原空間 dim}=2$$

$$\mathbf{m} = \frac{1}{4} \sum \mathbf{x}_i = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} \mathbf{C} &= \frac{1}{n-1} \sum_{i=1}^4 (\mathbf{x}_i - \mathbf{m})(\mathbf{x}_i - \mathbf{m})^T \\ &= \frac{1}{3} \left\{ \begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix} \right\} \\ &= \frac{1}{3} \begin{bmatrix} 10 & 10 \\ 10 & 10 \end{bmatrix} \end{aligned}$$

$$\det(\mathbf{C} - \lambda \mathbf{I}) = 0 \quad \left| \begin{bmatrix} \frac{10}{3} - \lambda & \frac{10}{3} \\ \frac{10}{3} & \frac{10}{3} - \lambda \end{bmatrix} \right| = \left(\frac{10}{3} - \lambda \right)^2 - \left(\frac{10}{3} \right)^2 = \frac{100}{9} - \frac{20}{3}\lambda + \lambda^2 - \frac{100}{9}$$

$$\lambda \left(\lambda - \frac{20}{3} \right) \Rightarrow \lambda = 0, \frac{20}{3}$$

$$(\mathbf{C} - \lambda) \mathbf{v} = 0$$

$$\begin{bmatrix} \frac{10}{3} - \frac{20}{3} & \frac{10}{3} \\ \frac{10}{3} & \frac{10}{3} - \frac{20}{3} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

$$\frac{1}{3} \begin{bmatrix} -10 & 10 \\ 10 & -10 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

$$\Rightarrow \begin{cases} \lambda = 0 & \mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \\ \lambda = \frac{20}{3} & \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \end{cases}$$

$$\begin{aligned} \mathbf{y} &= \mathbf{v}^T (\mathbf{x} - \mathbf{m}) \\ &= \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}^T (\mathbf{x} - \mathbf{m}) \\ &= \boxed{\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}} (\mathbf{x} - 0) = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \mathbf{x} \end{aligned}$$

→ 要作 normalized

$$\begin{bmatrix} \frac{10}{3} & \frac{10}{3} \\ \frac{10}{3} & -\frac{10}{3} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

$v_1 - v_2 = 0 \quad v_1 = v_2$
 $-v_1 + v_2 = 0 \quad -v_1 + v_2 = 0$
 $\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} v_1 \\ v_1 \end{bmatrix} = v_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$$\mathbf{z}_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{4}{\sqrt{2}} \\ 0 \end{bmatrix} = \begin{bmatrix} 2\sqrt{2} \\ 0 \end{bmatrix}$$

$$\mathbf{z}_2 = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{2}{\sqrt{2}} \\ 0 \end{bmatrix} = \begin{bmatrix} \sqrt{2} \\ 0 \end{bmatrix}$$

$$\mathbf{z}_3 = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ 0 \end{bmatrix} \quad \mathbf{z}_4 = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}$$

principal component analysis

→ principal component analysis



Remarks on PCA

$\vec{x}_i \in \mathbb{R}^d, i=1 \dots n \Rightarrow$ solve $L\vec{v} = \lambda\vec{v}$ for $\vec{v}_1, \vec{v}_2, \dots$ then the transformation
 find loading n eigenvector $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ $\Rightarrow L = [\vec{v}_1 \ \vec{v}_2 \ \dots \vec{v}_n]_{d \times n}$
 較大的特徵值

$\Rightarrow$ finally, $\vec{x} \in \mathbb{R}^d$. test data

new feature vector $\vec{z}$:

$$\vec{z} = L^T (\vec{x} - \vec{m})$$

$$\therefore \vec{z} \in \mathbb{R}^n, n \times d$$

Two critical issues :

1. 在降維的過程中，如何選擇特徵向量的個數當作投影軸來得到新的向量？ $\rightarrow$ **best eigenvector number** = ?

Solution $\hat{z}$ "optimal value of n is experimentally determined

2. 新的低維度特徵向量和原有的高維度特徵向量之間有何關聯？討論這個關聯有何意義？ $\rightarrow$ **interpretation of the new features** in the lower-dimensional vector = ?



Issue 1 : PCA最佳特徵向量選擇

(1) Cumulative variance approach (累積變異法)

- One popular method is based on the percentage of cumulative variance explained. This method determines the number of components by predefining a threshold on the cumulative variance.
- Definition. Variance explained and cumulative variance.**

(2) Wrapper approach → 直接利用分類率選擇

- The best number of chosen eigenvectors n is experimentally determined within the range between 1 and the maximum number of eigenvectors d . The best n is the one which results in the highest k -fold cross-validation classification accuracy.

$$\text{Ex } \mathbb{R}^3 \Rightarrow CV = \lambda V \text{ 得到 } \lambda_1, \lambda_2$$

$$\text{Assume } \lambda_1 = 1, \lambda_2 = 0.8, \lambda_3 = 0.2$$

$$\text{Def: Variance explained} = \frac{\lambda_i}{\sum_i \lambda_i}$$

$$\text{Ex variance explained } (\lambda_1) = \frac{1}{2} = 0.5 = 50\%$$

$$\text{Def Cumulative variance (CV)}$$

$$CV(\lambda_k) = \frac{\sum_{i=1}^k \lambda_i}{\sum_i \lambda_i}$$

$$\text{Ex } CV(\lambda_2) = \frac{1+0.8}{2} = \frac{1.8}{2} = 90\%$$



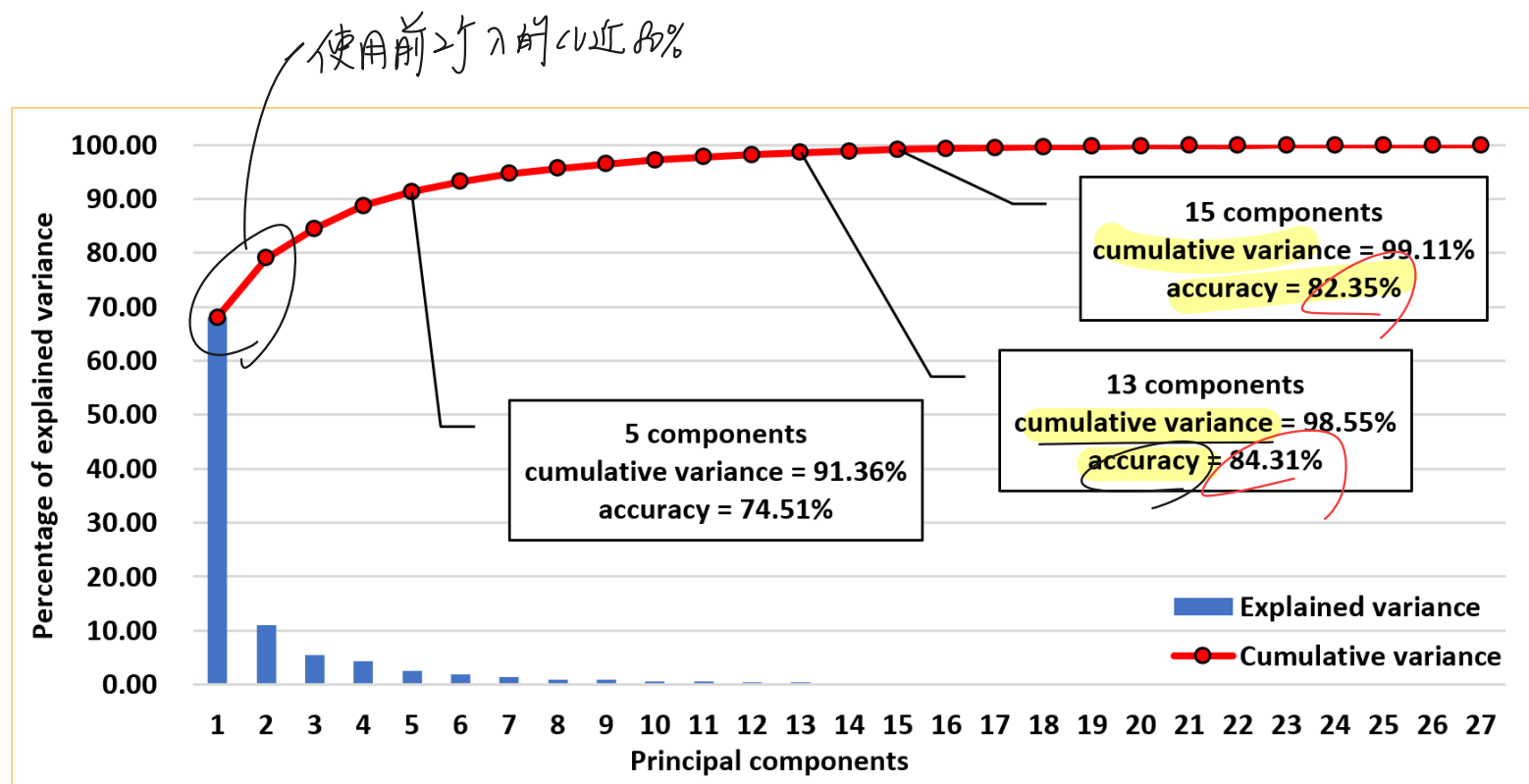
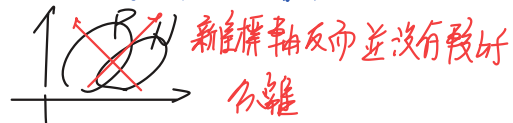
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Issue 1 : PCA最佳特徵向量選擇 – an example of cumulative variance method

1. If we determine the number of principal components by defining a cut-off threshold on the cumulative variance, 90% for example, the first 5 components explaining 91.36% of the variance would be selected.
2. However, the first 5 components gave only 74.51% cross-validation classification accuracy, lower than that provided by the first 13 ones.

3. **Advantage and weakness.** Although it is easy to implement, it could fail to find the best component combination which achieves the optimum classification performance.

* λ 越大 $\rightarrow$ 座標軸越好?



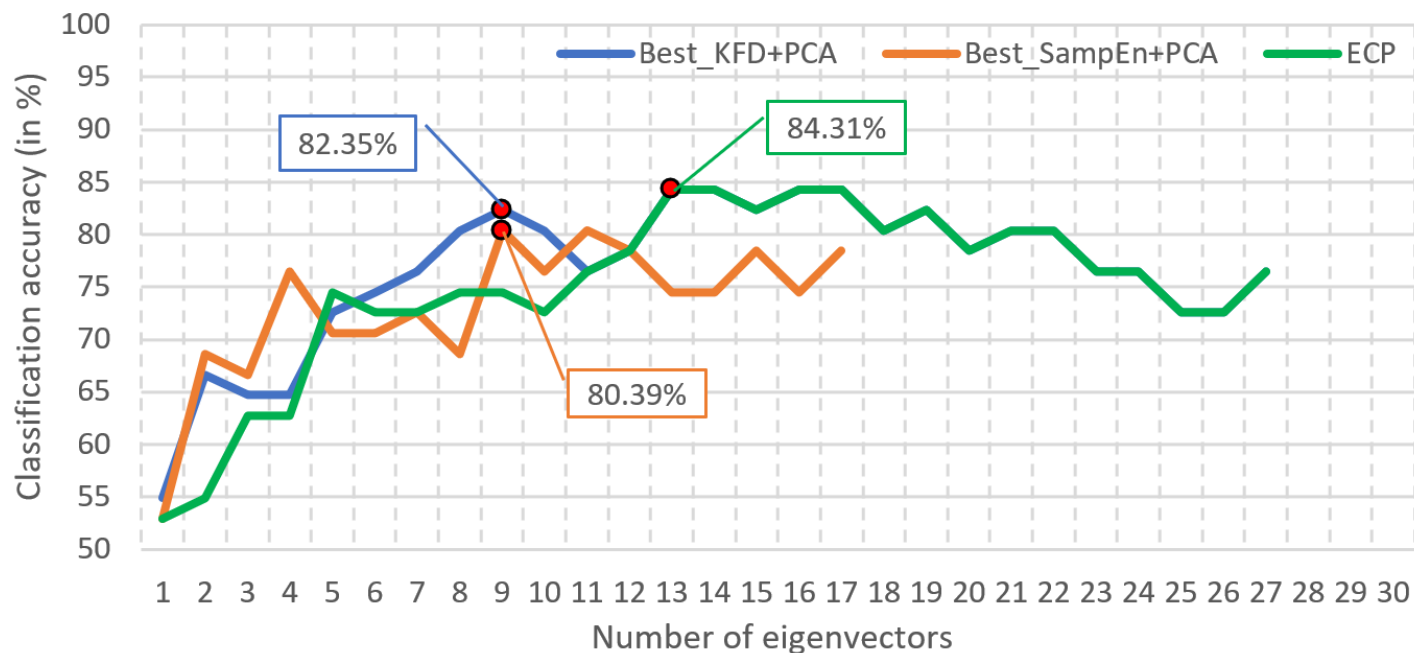
The blue bar and the red curve stand for the percentage of variance explained by each component (eigenvector) and the cumulative variance, respectively.



NessLab

Issue 1 : PCA最佳特徵向量選擇

– optimal eigenvector selection based on accuracy



Comparison of LDA classifier-based LOPO-CV classification accuracy among ECP, Best_SampEn+PCA, and Best_KFD+PCA features in different number of chosen eigenvectors. The maximum numbers of chosen eigenvectors for ECP, Best_SampEn+PCA, and Best_KFD+PCA are 27, 17, and 10, respectively. The numbers of the best PCA eigenvectors for **ECP**, Best_SampEn+PCA, and Best_KFD+PCA are **13**, 9, 9, respectively, where the highest accuracies (84.31%, 80.39%, 82.35%, respectively) are obtained for the three features.



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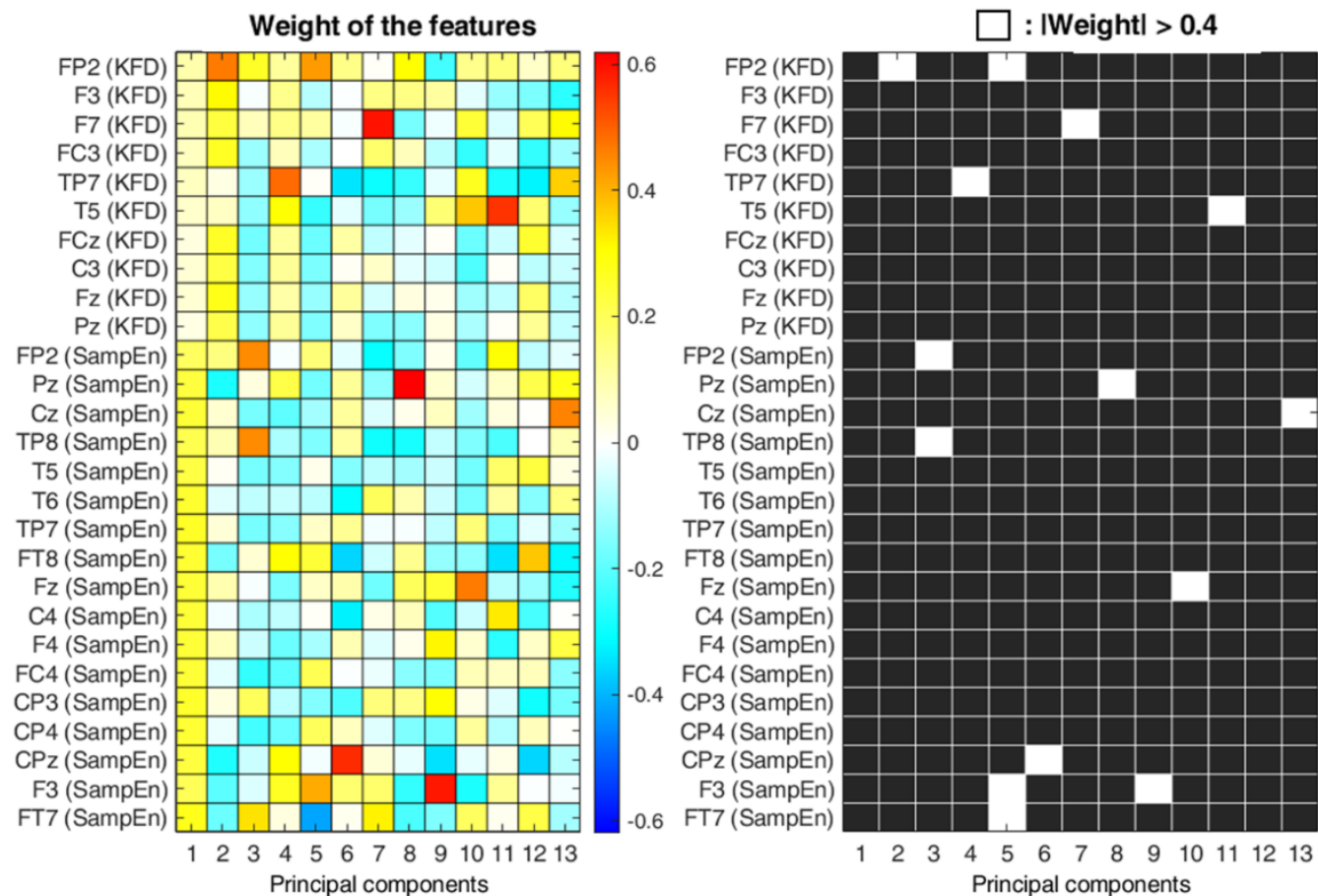
Issue 2 : Interpretation of the new features

手推例：



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Issue 2 : Interpretation of the new features



(Left) Weights of the 27 different features in the principal components, where the color represents the weight value. (Right) The features having weight value higher than 0.4 or lower than -0.4.



Limitations of PCA (1/2)

1. Due to its nature of unsupervised learning, the leading eigenvectors may might not be the optimum for classification.



Limitations of PCA (2/2)

2. Due to its nature of linearity, its performance drops when dealing with nonlinear data distributions.





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Kernel PCA

Basic Idea



Kernel PCA - formulation

Given a training set $\{\mathbf{x}_i \in R^n\}_{i=1}^M$, kernel PCA maps all the training data into a higher-dimensional feature space F via a nonlinear mapping $\varphi: \mathbf{x}_i \in R^n \rightarrow \varphi(\mathbf{x}_i) \in F$, where the mapped data centered to have zero mean $\sum_{i=1}^M \varphi(\mathbf{x}_i) = \mathbf{0}$.

Then, diagonalizes the covariance of the $\varphi(\mathbf{x}_i)$'s as

$$\mathbf{C}\mathbf{v} = \lambda\mathbf{v},$$

where $\mathbf{C} = 1/M \sum_{i=1}^M \varphi(\mathbf{x}_i)\varphi(\mathbf{x}_i)^T$ is data covariance, and $\mathbf{v}$ is an eigenvector associated with a nonzero eigenvalue λ .



Kernel PCA (cont.)

All solutions $\mathbf{v}$ lie within the span of the mapped data associated with spanning coefficients $a_i, i = 1, \dots, M$, *i.e.*,

$$\mathbf{v} = \sum_{i=1}^M a_i \varphi(\mathbf{x}_i)$$

Solving the problem $\mathbf{C}\mathbf{v} = \lambda\mathbf{v}$ is equivalent to solving the eigenvalue problem as

$$M\lambda\mathbf{a} = \mathbf{K}\mathbf{a},$$

where $\mathbf{K}$ is the $M \times M$ kernel matrix in which the entries are

$$K_{ij} = K(\mathbf{x}_i, \mathbf{x}_j) = \varphi(\mathbf{x}_i) \cdot \varphi(\mathbf{x}_j),$$

and $\mathbf{a}_k = (a_{k1}, \dots, a_{kM})^T$ are the eigenvectors of the kernel matrix subject to the normalization constraint $\|\mathbf{a}_k\|^2 = 1/\lambda_k, \forall k = 1, \dots, M$.



Kernel PCA (cont.)

Testing stage

The first d leading eigenvectors $\mathbf{v}'$ s are chosen as the projection axes. For a test data $\mathbf{x}$, projection of its image $\varphi(\mathbf{x})$ onto the p th eigenvector $\mathbf{v}_p$ is calculated as

$$y_p = \varphi(\mathbf{x}) \cdot \mathbf{v}_p = \sum_{i=1}^M a_{pi} \varphi(\mathbf{x}_i) \cdot \varphi(\mathbf{x}) = \sum_{i=1}^M a_{pi} K(\mathbf{x}', \mathbf{x}_i), \quad p = 1, \dots, d,$$

The nonlinear principal components corresponding to φ form a d -dimensional vector :

$$\mathbf{y} = (y_1, \dots, y_d)^T, \quad 1 \leq d \leq M.$$